

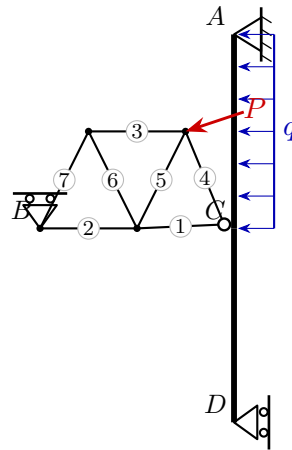
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Exam **Group A**

Date: 26.06.2026

Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

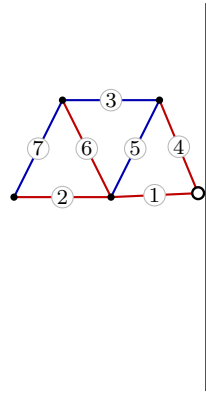
$$A : R_x = 14.25, R_y = -3.75 \quad D : R_x = 9.75, R_y = 0 \quad B : R_x = 0, R_y = 8.75$$

$$C : H_x = -15, H_y = 3.75$$

3. Truss member forces

Member	Force S [kN]	Type
1	13.12	tension
2	4.38	tension
3	-8.75	compression
4	4.19	tension
5	-9.78	compression
6	9.78	tension
7	-9.78	compression

red: tension (+) blue: compression (–)



4. Section-force diagrams of the beams

